

DHRUVKUMAR PATEL

Software Developer

+91-9328132446 | iamdhruv1563@gmail.com | linkedin.com/in/stack-dhruv | github.com/stack-dhruv

EDUCATION

IIIT DELHI

M.Tech in Computer Science and Engineering

Cumulative GPA: 8.0/10.0

Delhi, IN

Jul 2024 - Jun 2026

LDRP ITR

B.Tech in Information Technology

Cumulative GPA: 8.14/10.0

Gandhinagar, IN

Aug 2020 - May 2024

WORK EXPERIENCE

MIDAS: Multimodal Digital Media Analysis Lab

Graduate Researcher

IIIT-Delhi, IN

Dec 2024 – Present

- Developed and implemented highly scalable **REST APIs** using **FastAPI** & **Celery** for **asynchronous batch processing** of image inputs, achieving a **35% reduction in latency** for **automated anomaly detection** & analytics.
- Improved request throughput by **100+** requests/second by deploying a **GPU-backed containerized inference backend**.

Ishitva Robotics Systems Pvt. Ltd.

Junior SDE Intern

Ahmedabad, IN

Jan 2024 – Jun 2026

- Built a robust **synthetic data generation pipeline** with **OpenCV (C++)** featuring **10+ augmentation strategies** and real-time monitoring, enabling **5x class balance improvement**, **reducing validation error by 12%** on downstream tasks.
- Engineered a **modular Python utility** for efficient **batch data orchestration**, streamlining large image dataset organization through **metric learning and clustering**, significantly **reducing manual efforts by 80%**.

PROJECTS

DEATHSTARBENCH – SOCIAL NETWORK

[Github Link](#) | Feb 2025 – Mar 2025

- Deployed and orchestrated a **microservices-based** social network using **Docker**, & **Kubernetes**, and migrated from local clusters to **Google Cloud Platform** for scalability and observability.
- Conducted **throughput benchmarking** & **observability** with **Pixie**, **Jaeger**, and custom load testing to optimize resource allocation, inter-service communication, and **improved latency by 5%**.

WIFI COMMUNICATION PROTOCOL SIMULATION

[Github Link](#) | Nov 2024 – Dec 2024

- Simulated WiFi-4 (CSMA/CA), WiFi-5 (MU-MIMO), and WiFi-6 (OFDMA) protocols in **C++**, modeling user-AP communication, interference handling, and parallel channel access.
- Evaluated protocol capabilities across diverse network setups, applied **OOP principles for modular design** and automated builds using Makefile, and reported **10% higher throughput** with WiFi6 compared to WiFi4.

ADVANCED ANPR & FR (National Hackathon organized by MoE, India)

[Github Link](#) | Feb 2023 – Aug 2023

- Assembled **modular FastAPI** backend inference system supporting asynchronous OCR, identity verification, and image classification with **sub-200-millisecond latency** for edge deployment.
- Constructed a **decoupled pipeline** for camera-based license plate and face verification, integrating rule-based classification with API-driven client-server interactions.

PUBLICATIONS & RESEARCH

- “Unlocking Enigmatic Pathways: Empowering Student Dropout Analysis with Machine Learning and Energizing Holistic Investigation”, *2024 IEEE 9th International Conference for Convergence in Technology (I2CT)*
- (Under Review) “Modular Refinement of Small Language Models for Physics Reasoning via Reward-Driven Error Detection, Localization and Correction”, *The 40th Annual AAAI Conference on Artificial Intelligence, 2026*
- (Under Review) “Smart Traffic Surveillance: Deep Learning for Helmet Compliance and Vehicle Identification”, *The 40th Annual AAAI Conference on Artificial Intelligence, 2026*

ACTIVITIES

- Placement Coordinator – Placement Cell, IIIT Delhi** May 2025 – Present
Coordinated placements for 100+ M.Tech students and 10+ recruiters during 2025–26 hiring season.
- Executive Committee Member – TEDxIIITD** Dec 2024 – Apr 2025
Curated speakers and coordinated audience engagement for TEDxIIITD 2025 event with 500+ attendees.

ADDITIONAL

- CERTIFICATIONS:** AWS MLA-C01 (*Machine Learning Associate*), Microsoft Azure DP-100 (*Data Scientist Associate*)
- EXPERTISE AREA:** Software Engineering, Machine Learning, REST APIs, LLMs, Distributed Systems, NoSQL
- PROGRAMMING LANGUAGES:** C++, Python, SQL, JavaScript
- TOOLS & TECHNOLOGIES:** Git, GitHub, MySQL, Postman, FastAPI, GCP, AWS, Azure, Docker, Kubernetes
- ML & APIS:** PyTorch, SciKit-Learn, TensorFlow, OpenAI API, Gemini API, OpenCV, Hugging Face Transformers